

Неравенство Прекопы–Лейндлера

Homework

1 сентября 2026 года

1. (Prékopa 1973) Let $F: \mathbb{R}^n \times \mathbb{R}^m \rightarrow [0, +\infty)$ be log-concave and integrable in the second variable. Prove that the marginal $\varphi(x) = \int_{\mathbb{R}^m} F(x, y) dy$ is log-concave.
2. Deduce: (a) the convolution of two log-concave functions on $\mathbb{R}^n$ is log-concave; (b) the sum of independent random vectors with log-concave densities has a log-concave density; (c) for a log-concave probability density f on $\mathbb{R}$ the tail $\bar{F}(t) = \int_t^\infty f$ is log-concave.
3. Let γ_n be the standard Gaussian measure. Prove that $\gamma_n(\lambda A + (1 - \lambda)B) \geq \gamma_n(A)^\lambda \gamma_n(B)^{1-\lambda}$, and that for a convex K and a vector v the function $t \mapsto \gamma_n(K + tv)$ is log-concave; deduce $\gamma_n(K + v)\gamma_n(K - v) \leq \gamma_n(K)^2$.
4. (Brunn) For a convex body $K \subseteq \mathbb{R}^n$ let $s(t) = |K \cap \{x_n = t\}|_{n-1}$. Prove that $s^{1/(n-1)}$ is concave on its support; deduce that for $K = -K$ the central section is the largest, and give an example showing that symmetry is needed.
5. (Grünbaum 1960) Let K be a convex body with barycentre at the origin and let $H = \{x : \langle x, u \rangle \geq 0\}$. Prove $|K \cap H| \geq \left(\frac{n}{n+1}\right)^n |K| > |K|/e$. (*Hint: by Problem 4 the profile of the sections may be replaced by a linear one, i.e. the body by a cone.*)
6. Prove that the following functions are log-concave: the indicator of a convex set; $e^{-\|x\|}$ for an arbitrary norm; the Gaussian density; the Gamma density $x^{a-1}e^{-x}\mathbf{1}_{x>0}$ for $a \geq 1$. Show also that $e^{-|x|^p}$ on $\mathbb{R}$ is log-concave if and only if $p \geq 1$.

7. Deduce the weighted AM–GM inequality from Theorem 1 in dimension one applied to indicators of intervals, and describe the equality case in terms of f and g .
8. (Proposed problems, №266, M. Tetiva) Let $a \geq b \geq c > 0$ with $b^2 \geq ac$. Show that $\varphi(x) = \left(\frac{a^x + b^x + c^x}{3}\right)^{1/x}$ is logarithmically concave on $(0, \infty)$.
9. Let X be a real random variable with a log-concave density, $\bar{F}(t) = \mathbb{P}(X \geq t)$. Prove that $\bar{F}(s+t)\bar{F}(0) \leq \bar{F}(s)\bar{F}(t)$ for $s, t \geq 0$; show that for the exponential distribution this is an equality, and deduce that a log-concave density has finite moments of every order.
10. Let $K \subseteq \mathbb{R}^n$ be a convex body and f log-concave. Prove that $\Phi(x) = \int_K f(x+y) dy$ is log-concave; deduce that $x \mapsto \gamma_n(K+x)$ is log-concave and attains its maximum.